



Breach Series

Part 2: Breach Trace

Centre For Cybersecurity Institute

Project Breach Trace

Second part of the breach series under the SOC track simulates the monitoring and detection of the tactics, techniques, and procedures (TTPs) of a threat actor. This project challenges you to use SMB and Pass-the-Hash techniques to validate that the organization's SOC infrastructure can detect the malicious behaviour.

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Project Structure:

Scope 1

1. Add on to the PT script (.sh) that gets key information from the user (IP Address, usernames etc.) and let the user choose the following:
 - 1.1. Enumerates open smb shares on the windows client or Domain Controller,
 - 1.2. Dump hashes using impacket,
 - 1.3. Perform a DDOS attack using hping3 on the windows client.
2. Alerting and Monitoring:
 - 2.1. Create alert rules to detect ssh bruteforce, smb attacks, pass the hash attack, hping3 attack.
 - 2.2. Uses the hashes to perform a pass the hash attack manually to access the windows client and/or Domain Controller
 - 2.3. Configure Kibana dashboards to visualize real-time related security events and alerts.
3. Testing, Validation and Logging
 - 3.1. Execute penetration testing scripts against infrastructure.
 - 3.2. Monitor the Elastic Stack for generated alerts and security events.
 - 3.3. Validate the effectiveness of the alerting and monitoring system.
 - 3.4. On attack selection, save it into a log file in /var/log
 - 3.5. The log should hold the kind of attack, time of execution, and IP addresses
4. Documentation and Reporting:
 - 4.1. Document the setup process, including configurations and settings.
 - 4.2. Create a report detailing the findings, including detected security events and alerts.

General Tools available: nmap, NSEs, searchsploit, Metasploit, msfvenom, msfconsole, hydra, case scripting, find, dir, certutil, smbclient, impacket etc.

Project Deliverables:

- Record the execution of the .sh to show how the script work and alerts generated into an .mp4 format.
- Use comments in your code to explain what you did.
- If you are using code from the internet, add credit and links
- Put the script (.sh), .mp4 and pdf report into a Zip File.
- Upload the .zip and recording to drive
- File Name: UNIT.STUDENT NUMBER (e.g. CCK1_240101.s5.zip, CCK1_240101.s5.sh, CCK1_240101.s5.pdf etc.)
- Failure to do so will be taken as no submission and a 0 grade for the project
- Upload the project to your individual google drive folder
- Last Submission: 23:59 of the project submission deadline. Folder will be denied access thereafter.